

Linea Forced Transactions Audit Summary

Prepared by Cyfrin

Version 2.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea Forced Transactions. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated June 18, 2026, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Linea Forced Transactions code provided by Linea. The findings consist of 1 Low severity issue with the remainder being informational and gas optimizations.

Summary

Project Name	Linea Forced Transactions
Repository	linea-monorepo
Commit	990f688e9576...
Fix Commit	053093da2e7a...
Audit Timeline	Feb 2nd - Feb 6th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	1	1	0
Informational	6	5	1
Gas Optimizations	3	3	0
Total	10	9	1

6 Audit Scope

The audit scope was limited to Forced Transaction changes in PR897 (see the smart contract related changes at https://github.com/Consensys/linea-monorepo/pull/897/changes?mode=virtualization):

```
// new files
contracts/src/libraries/FinalizedStateHashing.sol
contracts/src/rollup/forcedTransactions/AddressFilter.sol
contracts/src/rollup/forcedTransactions/ForcedTransactionGateway.sol

// changes to existing files
contracts/src/rollup/LineaRollup.sol
contracts/src/rollup/LineaRollupBase.sol
contracts/src/rollup/LivenessRecovery.sol
```

7 Deployment Bytecode Verification

All verifications below used the same repository preparation, run at the stated commit:

```
gh repo clone Consensys/linea-monorepo
cd linea-monorepo
git checkout <commit>
pnpm install --ignore-scripts
cd node_modules/.pnpm/c-kzg@4.1.0/node_modules/c-kzg && npx node-gyp rebuild
cd contracts
npx hardhat test
```

Verification 1. At the conclusion of the audit, after all mitigations were concluded, Cyfrin was asked to verify that at commit 70dde7 (70dde763052985302cfaf4ea8e98010d11278ea1) the locally compiled bytecode matched the expected bytecode listed at that commit. The locally compiled artifacts for `ForcedTransactionGateway`, `AddressFilter`, `LineaRollup` and `Validium` were diffed against the expected artifacts in `contracts/deployments/bytecode/2026-02-20`, and the `immutableReferences` of `ForcedTransactionGateway` were extracted from the Hardhat `build-info` file referenced by its dbg file and checked.

Verification 2. After the forced transactions code was merged to `main` at commit 4ddbeba (4ddbeba806f1a31719539cd507d7bde29e4b481d), Cyfrin was asked to verify that the build output in `contracts/deployments/bytecode/2026-02-20` still matches the locally compiled code, allowing for the additional `immutableReferences` block appended to the deployments JSONs after the Hardhat compile:

- `AddressFilter`, `LineaRollup` and `Validium` are byte-identical between the locally compiled artifacts and the expected artifacts.
- For `ForcedTransactionGateway`, the only difference is the additional `immutableReferences` block appended to the deployments JSON. The locally compiled `immutableReferences` (extracted with the command below from the Hardhat `build-info` file referenced by `ForcedTransactionGateway.dbg.json`) use different top-level AST identifiers than those embedded in the expected deployments JSON (AST IDs depend on the exact set of source files passed to the compiler), but the structure (the number of immutables, and the `length` and `start` offsets of every entry) is identical:

```
jq '.output.contracts."src/rollup/forcedTransactions/ForcedTransactionGateway.sol"
  .ForcedTransactionGateway.evm.deployedBytecode.immutableReferences' \
  build/build-info/<hash>.json
```

```
Local AST ID -> Deployments AST ID: entries (length @ start)
33772 -> 33169: 32 @ 1031, 32 @ 2307, 32 @ 3796
33775 -> 33172: 32 @ 610, 32 @ 2712
33778 -> 33175: 32 @ 1133, 32 @ 3571
33781 -> 33178: 32 @ 949, 32 @ 3604
33784 -> 33181: 32 @ 1082, 32 @ 1795
33787 -> 33184: 32 @ 750, 32 @ 1889
33790 -> 33187: 32 @ 338, 32 @ 3672
33794 -> 33191: 32 @ 455, 32 @ 3117, 32 @ 3406
```

Every immutable placeholder slot in the locally compiled deployed bytecode coincides exactly with an immutable placeholder slot in the expected deployed bytecode, confirming that the build output in that folder still matches the compiled code at commit 4ddbeba.

Verification 3. While deploying to their Devnet, Consensys found that the `LineaRollup` deployed bytecode exceeded the EIP-170 contract size limit and recompiled it with the optimizer `runs` setting lowered from `10_000` to `200` (the override applies only to `src/rollup/LineaRollup.sol`; `AddressFilter`, `ForcedTransactionGateway` and `Validium` remain at `runs: 10_000`). Cyfrin repeated Verification 2 at commit 6d0c5d4 (6d0c5d481a7b0e2a1e5ad829890f4cab8032cb66) against the expected bytecode at that commit with the same results: `AddressFilter`, `LineaRollup` and `Validium` are byte-identical, and the `ForcedTransactionGateway` `immutableReferences` have the identical structure shown above, confirming that the build output still matches the compiled code after the optimizer-runs change.

Verification 4. In May 2026 the `linea-monorepo` repository was migrated to the Linux Foundation Decentralized

Trust (LFDT) and renamed; it now lives at `LFDT-Lineth/lineth-monorepo`. Cyfrin confirmed that commits 70dde7, 4ddbeba and 6d0c5d4 are all present in the new repository with their original contents, and that the expected artifacts in `contracts/deployments/bytecode/2026-02-20` are byte-identical between commit 6d0c5d4 and the current `main` branch.

After deployment to Ethereum mainnet, Cyfrin verified that the on-chain runtime bytecode of the deployed contracts matches the expected bytecode stored in `contracts/deployments/bytecode/2026-02-20`:

- `AddressFilter`: 0x526AE78F0103Ae73F05449ae30eb626C1003784E
- `LineaRollup` (implementation): 0x59290394dDC1cF84e671701A929710643c343530

The on-chain runtime bytecode was fetched at block `25344318` and compared with the `deployedBytecode` field of the expected artifacts:

```
cast code 0x526AE78F0103Ae73F05449ae30eb626C1003784E --rpc-url <mainnet-rpc> > AddressFilter.onchain
cast code 0x59290394dDC1cF84e671701A929710643c343530 --rpc-url <mainnet-rpc> > LineaRollup.onchain
diff <(jq -r .deployedBytecode AddressFilter.json) AddressFilter.onchain
diff <(jq -r .deployedBytecode LineaRollup.json) LineaRollup.onchain
```

The on-chain runtime bytecode of both contracts is byte-identical to the `deployedBytecode` of the expected artifacts, including the trailing metadata (CBOR) hash. This confirms the deployed contracts were compiled from the same sources with the same compiler version (`0.8.33`) and settings (EVM version `Osaka`; optimizer `runs = 10_000` for `AddressFilter`, 200 for `LineaRollup`) recorded in `contracts/deployments/bytecode/2026-02-20`. The keccak256 hashes of the on-chain runtime bytecode are:

```
AddressFilter
  0x9802e00fbcfe819915bec75ceb50f3ca1534fcef5a01ab7fc49109a217d7fb2e
LineaRollup
  0xf4830e7ba3afea4b3fb94af86fb63db9e5200ad1c7c9722c0b18a4d26fb7b58f
```

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence which aims to provide the “Ethereum experience” at scale and has been previously audited by Cyfrin.

This audit covered the new Forced Transaction functionality, which aims to provide a “processing guarantee” allowing users to submit L1 transactions which must be processed by the L2 sequencer within a specified block deadline.

10 Attestation

Cyfrin attests that it performed a security review of the Linea Forced Transactions code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-06-18-cyfrin-linea-forced-txns-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=11bf1646aeb5fa5d9b5868b5bb13949f4618e78d6610276f9e001a5d53cec00b
F=2026-06-18-cyfrin-linea-forced-txns-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.